Analysis of the Effect of Altitude on $PM_{2.5}$ Concentration Across Lombardy, Italy

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Executive Summary

This study investigates the relationship between altitude and $PM_{2.5}$ concentration in Lombardy, Italy, using a Bayesian hierarchical spatio-temporal model fitted to daily monitoring station data (2016–2021), aggregated to a cyclic monthly temporal index.

Four models with different combinations of random effects were compared using WAIC, NLSCPO, and MLIK. A full spatio-temporal model, incorporating a Matérn SPDE spatial effect and a cyclic first-order random walk temporal effect, was selected as the best-fitting specification.

Moderately informative priors were specified for all hyperparameters, with a sensitivity analysis confirming that posterior inference was robust to reasonable perturbations in prior specification.

Results showed strong evidence of a negative relationship between altitude and $PM_{2.5}$ concentration, where a one standard deviation increase in altitude ($\approx$ 214 metres) corresponds to a posterior mean reduction of approximately 14.8% in $PM_{2.5}$ concentration, with a 95% credible interval of (6.2%, 22.6%).

Model diagnostics revealed adequate log-normality of residuals, though consistent overprediction at high-fitted values and residual indexing autocorrelation indicates the model does not fully capture extreme $PM_{2.5}$ concentrations. A Continuous Ranked Probability Score (CRPS) of $\approx 9.02\mu g\, m^{-3}$ confirms moderate predictive performance; however, as the primary goal of the report is inferential rather than predictive, the estimated altitude effect remains well-supported within the modelling framework. Future work incorporating covariates such as livestock density and population density could reduce confounding and further strengthen the robustness of inference.

1 Introduction

Modelling pollution levels across altitude is important for understanding air quality dynamics. At higher elevations, lower atmospheric pressure and stronger winds enable pollutant dispersion [7]. In Lombardy, Northern Italy, intensive livestock farming and areas of high population density are concentrated at lower altitudes in the Po Valley, where surrounding Alpine and Apennine topography results in the accumulation of cold air and pollutants.

$PM_{2.5}$, particulate matter with an aerodynamic diameter less than 2.5 micrometres, is used as a measure of pollution due to its ability to penetrate deep into the respiratory system and its sensitivity to local emission sources [4]. This report investigates the relationship between altitude and $PM_{2.5}$ concentration across Lombardy, Italy using a Bayesian hierarchical spatio-temporal model. This framework allows us to estimate the effect of altitude while accounting for spatial dependence and monthly temporal variation.

The primary research question is: *What is the effect of altitude on $PM_{2.5}$ concentration after accounting for spatial and temporal dependence?*

2 Data Overview and Pre-Processing

Daily $PM_{2.5}$ concentrations ($\mu g\, m^{-3}$) for 2016–2021 were obtained from an open-access dataset on livestock and air quality in Lombardy, Italy [2]. The dataset consists of observations of $PM_{2.5}$ concentration and altitude (in metres) from multiple monitoring stations across Lombardy over a six-year period, resulting in a large spatio-temporal dataset suitable for hierarchical modelling.

Zero-valued $PM_{2.5}$ observations were treated as missing, as such values are likely to reflect measurement or recording error rather than true ambient conditions [1]. These, along with pre-existing `NA` values, were retained in modelling rather than excluded, as `R-INLA` omits missing observations from the likelihood while still producing posterior predictions at those locations.

To capture seasonal structure while maintaining computational tractability, time was aggregated to a cyclic monthly index ($t \in \{1, \ldots, 12\}$). This also aligns with expected seasonal variation in $PM_{2.5}$ concentrations. Altitude was standardised to improve numerical stability while ensuring regression coefficients are interpretable on a consistent scale.

3 Bayesian Model Specification

We fit a series of Bayesian hierarchical models of varying combinations of random effects. All models include an intercept and a fixed effect for altitude, differing only in whether they incorporate spatial and temporal random effects (Table 1). In doing so, we investigate whether accounting for residual spatial and temporal dependence in the data improves estimation of the altitude effect on $PM_{2.5}$ concentration.

Table 1: Summary of models fitted, differing in the random effects included.

Model	Spatial effect (u_s)	Temporal effect (m_t)
M0		
M1		✓
M2	✓	
M3	✓	✓

3.1 Observation Model

The distribution of observed $PM_{2.5}$ concentrations is illustrated in Figure 1.

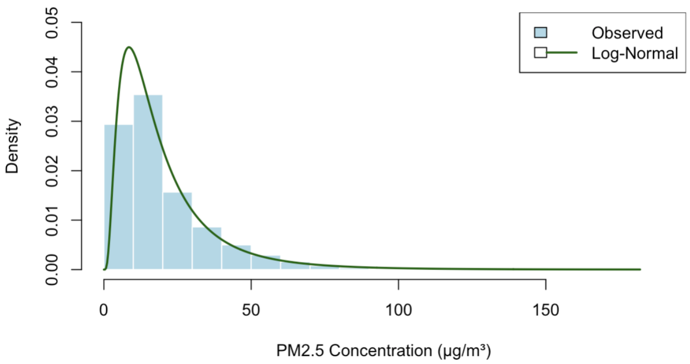

Figure 1: Histogram of observed $PM_{2.5}$ concentrations with an overlaid log-normal density fitted using maximum likelihood estimation (MLE).

$PM_{2.5}$ concentration is strictly positive and exhibits a right-skewed distribution, making the log-normal model a natural choice. This specification preserves interpretability, as covariate effects act multiplicatively on $PM_{2.5}$ on the original scale.

Let $P_{s,t}$ denote the $PM_{2.5}$ concentration at location s and time t, and let A_s denote the standardised altitude at location s. The full spatio-temporal model (M3) is

$$P_{s,t} \sim \text{LogNormal}(\eta_{s,t}, \sigma^2), \tag{1}$$

with the linear predictor defined as

$$\eta_{s,t} = \mu + \beta_1 A_s + m_t + u_s. \tag{2}$$

Here, μ is the overall mean log-concentration of $PM_{2.5}$, β_1 is the estimated altitude effect, u_s is a spatial random effect, and m_t is a cyclic temporal random effect defined at the monthly scale, repeating every 12 months.

3.2 Latent Process Model

The temporal random effect m_t is modelled as a first-order cyclic random walk,

$$m_t \sim \text{RW}_1(\sigma_m^2), \quad t = 1, \ldots, 12, \tag{3}$$

where σ_m^2 is the random walk variance. This captures smooth month-to-month variation in $PM_{2.5}$ across the seasonal cycle.

The spatial random effect u_s is modelled as a zero-mean Gaussian process with Matérn covariance,

$$u_s \sim \mathcal{GP}\big(0,\ C(\cdot\,;\, \sigma_u^2,\, \kappa)\big)\,, \tag{4}$$

where 0 denotes the zero-mean function, σ_u^2 is the marginal variance, κ is the spatial range, and the smoothness parameter ν is fixed at 1 under the Stochastic Partial Differential Equation (SPDE) approximation and is therefore not estimated.

3.3 Prior Model

The priors of our model were chosen to be moderately informative to reflect limited prior knowledge, ensuring they are vague enough to allow the data to dominate inference, yet still constrained enough to regularise estimation by discouraging implausible parameter values.

As such, the following priors were chosen for the fixed effects,

$$\mu \sim N(0, 1) \tag{5}$$
$$\beta_1 \sim N(0, 1) \tag{6}$$

These priors are centred at zero, implying no prior directional preference, and place most probability mass on moderate effect sizes on the log scale. This avoids implausibly large multiplicative effects on $PM_{2.5}$ concentration while remaining sufficiently weak to allow the data to dominate inference.

Additionally, a penalised complexity (PC) prior was chosen for the scale hyperparameter, σ, of the log-normal response, specified such that,

$$\mathbb{P}(\sigma > 1) = 0.01 \tag{7}$$

reflecting the belief that residual variability is present but unlikely to produce extreme dispersion on the log scale.

The priors on the random walk precision $(1/\sigma_m^2)$, and the precision $(1/\sigma_u^2)$ and range κ of the Matern Gaussian Process were all penalised complexity (PC) priors, with parameters determined by the following contraints.

$$\mathbb{P}(\sigma_m > 1) = 0.01 \tag{8}$$
$$\mathbb{P}(\sigma_u > 1) = 0.5 \tag{9}$$
$$\mathbb{P}(\kappa < 1) = 0.5 \tag{10}$$
$$\tag{11}$$

The constraint $\mathbb{P}(\sigma_m > 1) = 0.01$ reflects the belief that temporal variation in $PM_{2.5}$, after accounting for fixed effects, is unlikely to be extreme, as concentrations tend to change gradually over time instead of abruptly. The constraint $\mathbb{P}(\sigma_u > 1) = 0.5$ is less informative, suggesting more uncertainty about the magnitude of spatial variation, as $PM_{2.5}$ concentrations can vary considerably across regions depending on local topography. The constraint on the prior for the Matern range parameter (κ) was chosen to reflect the belief that the spatial random effect will act on a regional rather than hyper-local scale, with $\kappa = 1$ corresponding to a range parameter of approximately 100km after converting from degrees to kilometers.

4 Model Implementation

Direct computation with Gaussian processes such as 4 becomes computationally expensive at the scale of our dataset. To address this, the Stochastic Partial Differential Equation (SPDE) approximation of [6] is used, which represents the Matérn field as a sparse Gaussian Markov Random Field defined on a triangulated mesh over the study region (Figure 2). This renders computation tractable while preserving the spatial dependence structure of the Gaussian process.

All models were fitted in `R` using the `inlabru` package, which provides an efficient framework for SPDE-based models. Inference is performed using INLA, which avoids Markov Chain Monte Carlo (MCMC) sampling by employing deterministic approximations, and therefore does not require traditional convergence diagnostics.

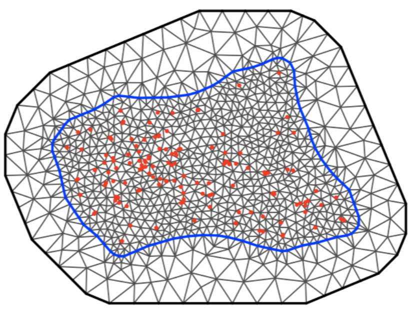

Figure 2: Triangulated mesh over Lombardy constructed from monitoring station locations (red points), used to compute the SPDE approximation of the Matérn spatial random effect.

4.1 Model Comparison

We assess the "goodness-of-fit" of all models specified for our analysis (M0–M3) using three scoring rules; Widely Applicable Information Criterion (WAIC), Negative Logarithm Sum of Conditional Predictive Ordinates (NLSCPO) and Marginal Log-Likelihood (MLIK). For WAIC and NLSCPO, a smaller value indicates a better predictive fit, while a less negative value for MLIK indicates stronger predictive fit.

WAIC penalises complexity, while NLSCPO evaluates the likelihood of each observation given all the other data points. MLIK assesses the likelihood of observing our data marginalised over all parameters of our model.

Table 2: Model comparison using the Widely Applicable Informative Criterion (WAIC), Negative Logarithm Sum of Conditional Predictive Ordinates (NLSCPO), and Marginal Log-Likelihood (MLIK).

Model	M0	M1	M2	M3
WAIC	797,554	762,638	790,805	754,463
NLSCPO	613,840	596,313	610,457	591,549
MLIK	-390,379	-373,735	-387,196	-369,327

For all three diagnostic metrics, the spatio-temporal model (M3) provides the best predictive performance among candidate models, due to obtaining lower WAIC and NLSCPO scores, and higher MLIK (see Table 2). This indicates that including both spatial and temporal random effects results in a model with better predictive fit on observed $PM_{2.5}$ concentration levels in Lombardy, Italy. Additionally, the temporal model (M1) outperforms the spatial model (M2) across all three metrics, suggesting that temporal structure explains more residual variation than spatial structure when included separately.

As a result, the spatio-temporal model (M3) is selected for further analysis.

4.2 Sensitivity Analysis

To assess the impact of prior specification on posterior inference, we conducted a sensitivity analysis using weakly, moderately, and strongly informative priors (see Appendix B). This evaluates the robustness of results to reasonable variations in prior assumptions. Posterior distributions under each specification for model M3 are shown in Figure 3.

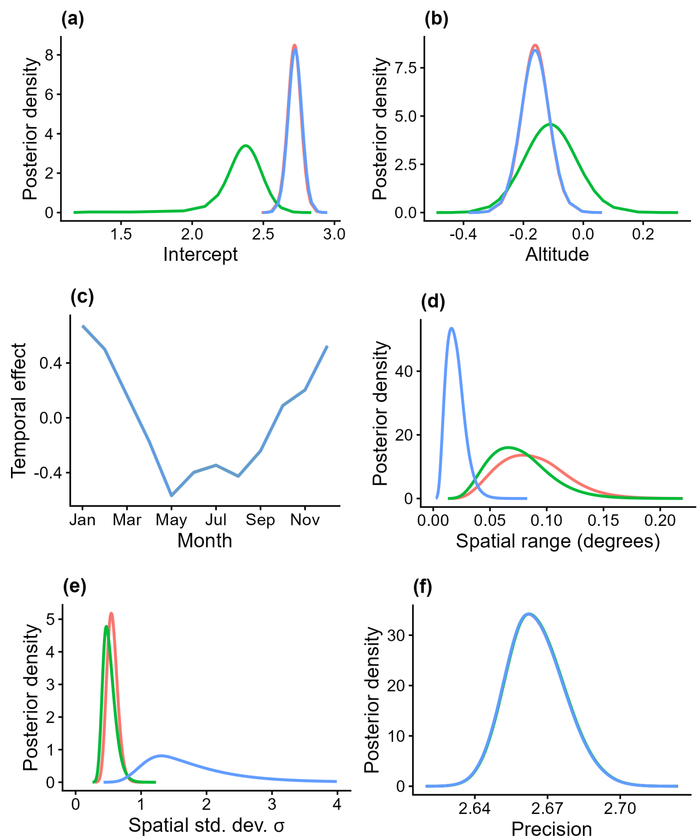

Figure 3: Posterior sensitivity to prior specification for the fixed effects: (a) intercept μ, (b) altitude coefficient β_1; (c) posterior mean of the RW1 temporal random effect across months; and posterior marginals of the Matérn SPDE hyperparameters: (d) spatial range κ, (e) spatial standard deviation σ_u; and (f) the log-normal likelihood precision ($1/\sigma^2$). Blue denotes weakly informative priors, Red denotes moderately informative priors, and Green denotes strongly informative priors.

Zero-mean normal priors with three precision specifications were investigated for the fixed effects. The weakly and moderately informative priors produced near-identical posterior marginals, indicating that inference for fixed effects is largely data-driven across a reasonable range of prior specifications. The strongly informative prior produced noticeably different posteriors, indicating that such priors are overly restrictive given the data (Fig. 3(a),(b)).

PC priors placed on the log-normal likelihood precision and the temporal effect both produced essentially identical posteriors across all three specifications (Fig. 3(c),(f)), suggesting both components are robust to prior specification.

Varying the prior on the spatial range and standard deviation had a more notable impact on posterior inference. The weakly informative range prior placed substantial probability on short distances, resulting in hyper-local spatial effects inconsistent with the scale of the study domain. Both the moderately and strongly informative priors successfully avoided this, producing broadly consistent posteriors for both parameters (Fig. 3(d),(e)).

Overall, the sensitivity analysis supports the use of moderately informative priors across all components, as they consistently avoided instability introduced by overly diffuse priors while being less restrictive than the strongly informative alternatives, which allows the data to drive posterior inference. For this reason, the moderately informative priors will continue to be used in subsequent analysis.

4.3 Evaluation of Selected Model

We employ Bayesian studentised residuals to assess the distributional assumptions of the selected spatio-temporal model (M3) (Figure 4).

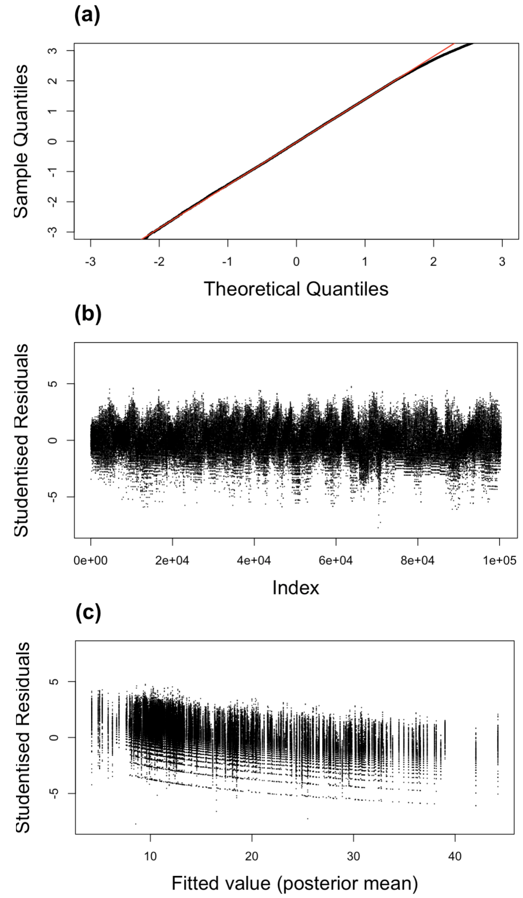

Figure 4: Residual diagnostics for the final spatio-temporal model (M3). (a) log-normal Q-Q plot of Bayesian studentised residuals, (b) Bayesian studentised residuals against observation index, (c) Bayesian studentised residuals against posterior mean fitted values.

The Quantile-Quantile (Q-Q) plot (Figure 4(a)) shows that residuals closely follow the theoretical quantiles expected under the log-normal model, indicating that the assumed log-normality residual structure is sufficiently satisfied, with only minor deviation in the upper tail.

Figure 4(b) reveals a cyclical pattern in the residuals when ordered by observation index, suggesting some residual temporal or sequential dependence not fully captured by the monthly random walk effect. This likely reflects submonthly temporal variation or station-level autocorrelation, suggesting finer temporal resolution could improve model fit.

Figure 4(c) diagnostic reveals a systematic negative trend in residuals as the posterior mean increases, with approximately constant spread, indicating some degree of bias in the model, with higher $PM_{2.5}$ concentrations being overestimated. This may reflect regularisation induced by the PC priors, or limitations of the log-normal specification in capturing extreme values.

While this does not invalidate inference on the fixed effects or random effect parameters, predictive performance at extreme locations should be interpreted cautiously.

Posterior predictive checks further highlight limitations in capturing the tails of the $PM_{2.5}$ distribution (Figure 5).

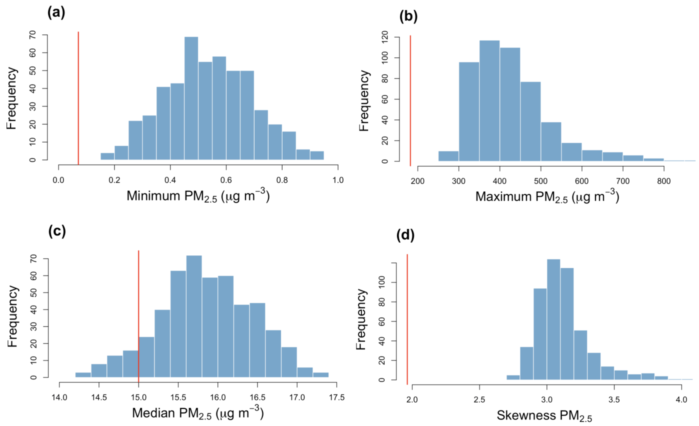

Figure 5: Histograms of summary statistics of 500 resamples of $PM_{2.5}$ data under model M3, with the value of the observed summary statistic marked in red: (a) minimum, (b) maximum, (c) median, (d) skewness.

By resampling 500 times from the fitted model, we compare the distribution of selected summary statistics against those of the observed data. These show the model consistently overpredicts both the minimum and maximum observed $PM_{2.5}$ values, and the resampled dataset exhibits substantially higher skewness than the observed data. This indicates the model does not adequately capture the extremes of the $PM_{2.5}$ concentration data. Median predictions show better agreement with observed values, though resampled medians still exceed the observed median with high probability.

The Continuous Ranked Probability Score (CRPS), a negatively-oriented proper scoring rule that jointly assesses the accuracy and calibration of the full predictive distribution [3], was computed as an additional measure of predictive performance. Our model produced a CRPS of $\approx 9.02\,\mu\text{g}\,\text{m}^{-3}$, which, given that most $PM_{2.5}$ values are within 0 to 50 $\mu\text{g}\,\text{m}^{-3}$ (as displayed in Figure 1), indicates moderate predictive accuracy relative to the scale of $PM_{2.5}$ concentrations.

5 Results

5.1 Effect of Altitude on $PM_{2.5}$ concentration

Altitude exhibits a negative relationship with $PM_{2.5}$ concentration, with a posterior mean estimate of -0.1603 and 95% credible interval of $(-0.2558, -0.0644)$ (see Table 3). As the credible interval is negative and excludes zero, there is strong evidence that $PM_{2.5}$ concentration decreases as altitude increases.

Table 3: Posterior estimates (mean, standard deviation (SD), and lower and upper bounds of 95% credible intervals) of fixed effects from the Bayesian spatio-temporal model.

Covariate	Mean	SD	2.5% CI	Median	97.5% CI
Intercept	2.7177	0.0493	2.6187	2.7182	2.8141
Altitude	-0.1603	0.0482	-0.2558	-0.1604	-0.0644

Posterior mean estimates of the model parameters will be used to summarise model effects. Under the Log-Normal model, after exponentiation these values correspond to multiplicative changes in the median of the response distribution (see Appendix A). At average altitude and with spatial and temporal effects set to 0, the baseline median $PM_{2.5}$ concentration is $\exp(2.7177) \approx 15.15\mu\text{g}\,\text{m}^{-3}$.

An increase in altitude of one standard deviation ($\approx$ 214m) corresponds to a reduction in median $PM_{2.5}$ concentration of $\approx$ 14.8% ($\exp(-0.1603) \approx 0.8520$) relative to this baseline median. The multiplicative factor has a 95% credible interval of (0.7743, 0.9376), corresponding to a reduction of between 6.2% and 22.6% ($\exp(-0.2558) \approx 0.7743$, $\exp(-0.0644) \approx 0.9376$), giving a considerable negative effect even at the bounds of the credible interval. This negative association is physically plausible, reflecting both reduced emission sources and greater atmospheric dispersion at higher altitudes.

5.2 Contribution of Spatial and Temporal Effects

The selected model M3 contains spatial and temporal random effects, which capture structured variation not explained by the effect of altitude, likely reflecting factors such as livestock density, industrial activity, or population density, all of which are known to affect $PM_{2.5}$ concentrations. [5, 10]. The fitted spatial random effect across our region of study is visualised in Figure 6.

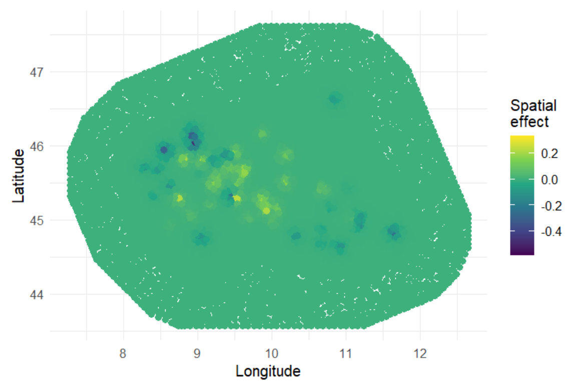

Figure 6: Spatial random effect over the study region.

Raw spatial effect values range from ≈ -0.4 to ≈ 0.3 on the log scale, and correspond to multiplicative effects of $\exp(-0.4) \approx 0.67$ to $\exp(0.3) \approx 1.35$, indicating that median $PM_{2.5}$ concentrations vary by up to roughly 33% below or 35% above the baseline level depending on location.

Additionally, we observe that higher altitude regions in the north and southwest of Lombardy, corresponding to the Alps and Apennines, do not show disproportionate spatial random effect values, suggesting that the spatial effect captures residual spatial variation rather than confounding the altitude effect. In fact, much of the positive spatial effect is concentrated around the Milan Metropolitan area, suggesting that the spatial effect may be capturing high population density and industrial activity in this area. The posterior mean Matérn range parameter was found to be approximately 10km, indicating that spatial dependence operates at a local rather than regional scale, further supporting that localised emission sources may drive much of the spatial structure in the data.

The random walk temporal effect captures seasonal variation in pollution levels without imposing a fixed seasonal form. The resulting random effect is displayed in Figure 7.

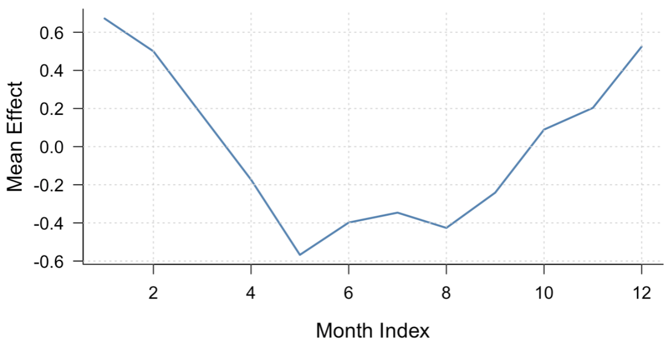

Figure 7: Mean temporal random effect across months of the year.

Since the temporal effect is constrained to have zero mean across months, positive values indicate above-average $PM_{2.5}$ levels, while negative values indicate below-average levels.

Winter months (October to March) show consistently positive values, while summer months (April to September) show consistently negative values. These observations are consistent with previous work on seasonal variation in pollution in this region, where winter peaks are attributed to increased home heating and reduced atmospheric dispersion [8].

The positive effect peaks in January, when the temporal effect results in a median approximately 80% higher than in the absence of a temporal effect ($\exp(0.6) \approx 1.822$), and reaches its minimum in May, corresponding to a reduction of approximately 45% ($\exp(-0.6) \approx 0.5488$).

Together, these results indicate that while altitude explains a systematic component of $PM_{2.5}$ variation, substantial additional structure arises from local spatial factors and strong seasonal effects.

6 Discussion

The results presented in Section 5 provide strong posterior evidence within the assumed model that $PM_{2.5}$ concentration decreases as altitude increases. The effect size is substantial, with a one standard deviation increase in altitude ($\approx$ 214m) corresponding to a reduction in median $PM_{2.5}$ concentration of approximately 14.8%. This is consistent with existing scientific work on pollution dynamics in Lombardy, which highlights the accumulation of pollutants from agriculture and heavy industry in the Po Valley [9].

The superior performance of the spatio-temporal model (M3) suggests the presence of meaningful spatial and temporal variability in $PM_{2.5}$ concentration, which are found to contribute to deviations of up to approximately 35% and 80% from the basline median $PM_{2.5}$ concentration, respectively (Section 5.2).

However, several limitations should be acknowledged. While the Q-Q plot confirms adequate fit under the log-normal assumption, mean studentised residuals were found to decrease with fitted values and exhibit autocorrelation when ordered by observation index. Additionally, posterior predictive plots revealed poor modelling of tail statistics. Furthermore, the CRPS of $\approx 9.02\,\mu\text{g}\,\text{m}^{-3}$ suggests a moderate, but not close predictive fit relative to the distribution of the data, indicating that M3 may not be well-suited for prediction purposes. While these diagnostics suggest that the posterior inference for the fixed and random effects should be interpreted cautiously, the evidence for the direction and magnitude of the effect of altitude is strong, and thus remains reasonably well supported by our analysis.

Further work could resolve these limitations by incorporating additional covariates into the analysis, such as livestock density, proximity to heavy industry, and population density. The inclusion of these covariates may reduce potential confounding effects and allow for more robust inference on the effect of altitude. Additionally, incorporating a space-time interaction would allow the model to capture location-specific temporal variation in $PM_{2.5}$, accounting for the fact that seasonal patterns may vary across locations.

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A Interpretation of coefficients with a Log-Normal response

Suppose the model under consideration has response $Y_i \sim \text{Lognormal}(\mu_i, \sigma^2)$, and has an intercept and only one linear covariate, $\mu_i = \beta_0 + \beta_1 x_i$. Then by standard properties of the Log-Normal distribution the median is given by

$$\text{Median}(Y_i) = e^{\beta_0 + \beta_1 x_i} = e^{\beta_0} e^{\beta_1 x_i}$$

so the second term acts as a multiplier of the first, varying with the covariate. Further, a one unit increase in x_i will result in a new median given by

$$\text{Median}(Y_i') = e^{\beta_0 + \beta_1 (x_i + 1)} = e^{\beta_1} e^{\beta_0} e^{\beta_1 x_i} = e^{\beta_1} \cdot \text{Median}(Y_i)$$

Similarly, appending a random effect r_j to the linear predictor $\mu_i = \beta_0 + \beta_1 x_i + r_j$ results in the median changing by

$$\text{Median}(Y_i'') = e^{\beta_0 + \beta_1 x_i + r_j} = e^{r_j} e^{\beta_0} e^{\beta_1 x_i} = e^{r_j} \text{Median}(Y_i)$$

so the exponential of the raw random effect value at the data point also acts as a multiplicative factor on the median response. It is worth noting that the mean response has the exact same multiplicative factors as the median response, but takes different values.

B Sensitivity Analysis Priors

Parameter	Prior Type	Weak	Moderate	Strong
μ (Intercept)	Gaussian	prec $= 0.001$	prec $= 1$	prec $= 10$
β_1 (Altitude)	Gaussian	prec $= 0.001$	prec $= 1$	prec $= 10$
σ_m (RW1 precision)	PC	$u = 0.1,\ \alpha = 0.01$	$u = 1,\ \alpha = 0.01$	$u = 10,\ \alpha = 0.01$
σ_u (Matérn SD)	PC	$\sigma_0 = 5,\ P = 0.5$	$\sigma_0 = 1,\ P = 0.5$	$\sigma_0 = 0.5,\ P = 0.5$
κ (Matérn range)	PC	$r_0 = 0.1,\ P = 0.5$	$r_0 = 1,\ P = 0.5$	$r_0 = 3,\ P = 0.9$
σ (likelihood precision)	PC	$u = 10,\ \alpha = 0.01$	$u = 1,\ \alpha = 0.01$	$u = 0.3,\ \alpha = 0.01$

Table 4: Prior specifications for model parameters under weak, moderate, and strong settings.